GO BEYOND	Pratt & Whitney Canada Corp. **Industrial Management Procedure**	IMP No.:07-01-01 REV: 2 DATE: 5 Jun 2025 PAGE: 1 OF 47
Procedure Title:	**Industrialization of new programs**	

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WHAT'S NEW IN THIS DOCUMENT - NATURE OF CHANGES

REV. 1 – Update IMP 07-01-01 document to PR deliverables, MRL S-Curve addition, Revised Industrialization RACI
REV. 2 – Update IMP 07-01-01 document to align with latest MMP220 rev13.

REVISION STATUS HISTORY - SEE PRIOR REVISIONS FOR DETAILS

Revision	Date	Edited by
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1 PURPOSE AND SCOPE

1.1. This procedure describes the requirements, processes, and responsibilities to the Operations Program Office (OPO) organization to convey the intent of MMP 220 RTX Lifecycle Program Management (RLPM) Process into new program requirements (new centerline and new program derivatives).

1.2. This procedure provides the MMP 220 requirements and the industrialization deliverable associated with the MRL target required to be met for each of the program phases. IMP 07-01-01 is also referencing P&WC forms & manual as listed in figure 2 below.

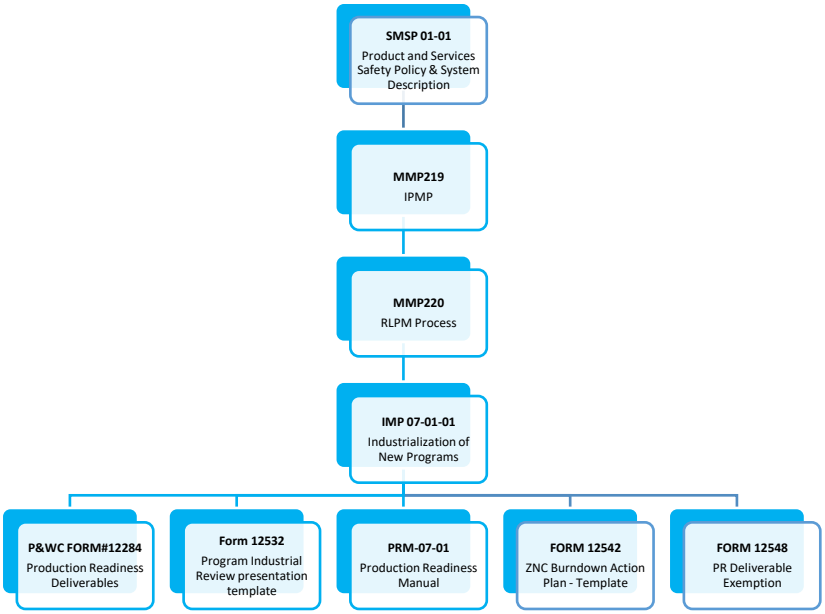

Figure1 – Link between MMP, IMP and P&WC Forms

1.3. This procedure provides:

A) The **Production Readiness (PR) deliverables**, synchronized to MMP 220 phases (RLPM Process);
B) The **Industrialization organization & responsibilities** for the OPO functions involved in industrialization activities: Program Industrialization Manager, Industrialization Project Manager, Partnership Industrialization Account Managers, Partnership Quality, Operation Program Engineer/Analyst, Product Cost Engineer/Analyst, Global Chemical Regulation Specialist, GT Focal, and GT Professional.

1.4. The maintenance and the next revisions of this procedure are to be jointly directed by the PMO and OPM organizations.

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2 ACRONYMS AND DEFINITIONS

2.1 ACRONYMS

AS	Account Specialist (SM)
AS*(with numbering)*	Aerospace Standard
BB	Bridge Buy
BPM	Business Process Management (SM)
CAR	Canadian Airworthiness Regulations
CDR	Critical Design Review
CERT	Certification
DFARS	Defense of Defense Federal Acquisition Regulation
DOD	Department of Defense
ECE	Engineering Cost Estimate
EIP	Entry Into Production
EIS	Enter/Entry Into Service
EV	Earned Value
EVP	Engine Validation Plan
FAR	Federal Acquisition Regulations
FETT	First Engine To Test
FG	Finished Goods
GCR	Global Chemical Regulations
GM&T	Global Manufacturing & Transitions
GT	Global Trade
IGR	Industrialization Gate Review
IMP	Industrial Management Procedure
IPM	Industrialization Project Manager
IPMP	Integrated Product Management Process
IPT	Integrated Product Team
KPI	Key Performance Indicator
LTB	Last/Lifetime Buy
MA	Manufacturing Approval
MA / ME	Manufacturing Analyst / Manufacturing Engineer
MFP	Member Focal Point
MMP	Management Manual Procedures
MRL	Manufacturing Readiness Level
MTE	Manufacturing & Tooling Engineering
NPI	New Product Introduction
NRE	Non-Recurring Expenses
NSA	New Supplier Assessment
NSD	New Supplier Development

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NSE	New Supplier Evaluation
NSI	New Supplier Introduction
OEM	Original Equipment Manufacturer (Aircraft – customer)
OPE / OPA	Operation Program Engineer/Analyst
OPM	Operations Program Management
O-PMT	Operations Program Management Team
O-PMMT	Operations Program Model Management Team
OPO	Operations Program Office
PBI	Power Business Intelligence (Microsoft)
PDR	Preliminary Design Review
PDRS	Product Definition Requirement Specifications
PIAM	Partnership Industrialization Account Manager
PIM	Program Industrialization Manager
PMMT	Program Model Management Team
PMO	Program Management Office
PMT	Program Management Team
POC	Parts of Concern
PO / SA	Purchase Order / Scheduling Agreement
PPAP	Production Part Approval Process
PQ	Partnership Quality
PQI	Partnership Quality & Industrialization
PR	Production Readiness
PRL	Producibility Readiness Level
PR-L2	Production Readiness – Level 2
PSSRB	Program Strategic Sourcing Review Board
QMSP	Quality Management System Procedures
QNLI	Quality Notification Line Item
RFP	Request For Proposal
R&T	Research & Technology
SDL	Supplier Development Lead
SFV	Strategic Financial Valuation
SHA	Supplier Health Assessment
SM	Supply Management
SQA MFI	Supplier Quality Assurance Material Flow Integrity
SQAR	Supplier Quality Assurance Representative
SMSP	Safety Management System Procedure
SSRB	Strategic Sourcing Review Board
STO	Support To Operations
TRL	Technology Readiness Level
TSO	Technical Standard Order
ZNC	Zero Non Conformity
4WBA	4-Week Build Ahead

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2.2 DEFINITIONS

Industrialization Industrialization structured approach is applied from RLPM Gate A to Gate F Industrialization's goal is to meet the operations program milestones and objectives for the development and serial production of a product. It includes operation planning, risk management and execution of the Production Readiness (PR) activities.

Production Readiness As part of the industrialization the Production Readiness is a process executed to ensure the desired PR Earned Value and Manufacturing Readiness Level (MRL) program targets are met to secure a flawless Entry Into Production (EIP). The "customer" of the PR process is the MMP 220 RLPM Process.

Derivative Incremental limited changes made to an existing product.

New Centerline A new product not being a derivative.

Producer P&WC Manufacturing, external suppliers, Partners, Engine Center Assembly & test or any entity providing a part of the product for the program.

Partners Entity providing goods or services to P&WC under which a NRE and/or Revenue and risk sharing agreement is made between the partner and the P&WC.

Product Goods sold by P&WC. Product may be a CAR 533 certified engine, a TSO C-77 (or E-TSO) Auxiliary Power Unit. Product may consist of aircraft components, systems and/or SW as part of the contract with the customer.

Producibility Producibility is the property of a part definition which allows: a stable, robust, repeatable and reliable manufacturing process that meets target cost; a predictable process lead-time; Manufacturing with Zero Recurring Non-Conformance (ZNC); Conformance with the design intent/requirements (fit-form-function). Ideally, the part definition should not depend on the expertise of a unique producer.

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3 REFERENCES

AS9145	Advanced Product Quality Planning & PPAP Requirements for the Aerospace & Defense Industry (IAQG)
CPW153	Management and Classification of Key Product Characteristics
CPW40	Packaging and Labelling
CSOP P-41	Customer Service Field Readiness (EIS) Process
DoD MRL	MRL Deskbook (Department of Defense v.2020)
DRM110	Traceability Practices (Serialization and Heat Identification)
EOP 089	Engineering Technology Management
EOP 103	Design Process and Design Reviews
EOP 257	Development Engineering Production Readiness Process
FCP-OPSH-EHS-21	Global Chemical Substances Management
IMP 07-05-06	Partnership – Work on Parts at PWC
IMP 08-03-02-Partner	Partnership – Control of Nonconforming Product
IMP 08-03-04	Partnership – Escape Management
IMP 08-03-05	Partnership – PPAP
MMP 060	Protection and Control of Intellectual Property
MMP 219	Integrated Product Management Process
MMP 220	RLPM Review Process
MMP 239	Process Certification
MMP 345	International Trade Compliance (GT)
MMP 348	Global Strategic Sourcing
MMP 375	International Trade Compliance
MMP 501	New Product Introduction Factory Standard Cost Management
MMP 519	Factory Standard Cost Procedure
MOP030241	Gestion du développement des technologies de fabrication
OPMP 3.1	Coordinate Manufacturing Approval
OPMP 3.10	Industrialization Pre-Passport 0 to P2
OPMP 3.11	Industrialization P2 up to Entry Into Service
OPMP 3.12	Legacy Program Management
OPMP 4.11	Production Hardware Management
PPP 02-01	Quality Systems
PPP 03-01	Contract Requirements Review and Communication
PPP 06-11	Dual Sourcing
PPP 06-14	Global Strategic Sourcing
PPP 06-15	Supplier Targeting and ~~RFx~~
PPP 07-12	Electronic Obsolescence Management Process
PPP 09-03	Vendor Supplied Material (VSM) Program
PPP 11-02	Gauges and Tooling Provided to Suppliers

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PPP 15-06	Transfer from Experimental to Production
PRM 07-01	Production Readiness Manual
PWAP 22.21	P&W Global Chemical Substances Program
PWC Form 10285	Process Robustness Index (PRI)
PWC Form 10865e	NPI to AS Transfer Checklist
PWC Form 11184	Produceability Readiness Level (PRL)
PWC Form 11737	Request for PMTS Approval
PWC Form 11847	NPI Operations Budget
PWC Form 11856	Part Transfer from MFP to QS Approval Sheet
PWC Form 11908	Dual Sourcing – SPF Decision Matrix
PWC Form 12284	Production Readiness Deliverables
PWC Form 12532	Program Industrial Review (PIR) presentation template
PWC Form 12542	ZNC Burndown Action Plan - Template
PWC Form 12548	PR Deliverable Exemption
QMSP 04-02	Document and Record Control
QMSP 04-02-01	Plans qualité (QP) et Plans de gestion de la collaboration (CMP)
QMSP 07-05-02	Production Part Approval Process
QPM	Pratt & Whitney Canada Quality Program Manual
SQOP 01-06	Part Marking and Tracking System (PMTS)
SQOP 01-08	New Part Quality Planning
WIM 292	Electronic Obsolescence Management
WIM 693	Producibility Readiness Level (PRL)
WIM 944	Manufacturing Readiness Level (MRL) Assessment
WIM 1048	Parts of Concern (MEPoC)

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4 REQUIREMENTS AND TASKS

4.1 GOVERNANCE STRUCTURE – PROGRAM AND INDUSTRIAL MANAGEMENT

4.1.1 IPMP Structure

Refer to MMP 219

4.1.1.1 OPO NPI Program Governance

OPO NPI program governance are held to review cross-program systemic issues, cross-pollenate best practices, train and inform industrialization leaders across the OPO organization as well as review critical program status. OPO NPI Program Governance are typically led by the PMO director.

4.2 PRODUCTION PLAN

In accordance with MMP220, it is required to prepare a comprehensive Production Plan for each program and is owned by Operations. The purpose of this plan is to provide a clear and overarching view of how the operations organization will implement Project Management Processes throughout the lifecycle of a program or project. This includes detailing the various phases of the program, all tailored to meet contractual requirements and program archetype.

Each section of the production plan should articulate strategic aspects, ensuring that readers can easily understand the rationale behind the plan. Additionally, the production plan must remain up to date beyond Gate D, consistently reflecting the Program's current status and strategic direction throughout the entire lifecycle of the product.

A "looking forward" production plan can be crated for any Program/Product at any time of its lifecycle to support RLPM.

The Production plan of each Program/Project needs to be maintained throughout the lifecycle of the product, regularly updated together with the other Management Plans and aligned with the respective Program plan. The updates to the production plan need to be discussed at the OPMT.

4.3 PRODUCTION READINESS PROCESS

4.3.1 Standards & Mandate

The Production Readiness is the process used as part of the governance for the industrialization of the new programs. At each MMP 220 RLPM gate, a set of exit criteria is defined to meet the industrialization requirements of the program and, to support the Gate decision. The PR deliverables are defined as part of P&WC form 12284.

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Table 1 below is aligning the relations between the Program Milestones, RLPM gates, IGR, MRL, TRL, Development Engineering gates, and APQP phases. Earned value targets listed in table 1 below are reflecting the P&WC form 12284 at the date of this IMP release date. If there is a difference between the PR EV target below and the one in the P&WC 12284 form, the later one prevails.

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MRL#	MRL definition (DOD)	IGR Objective	MMP220 Gate	EOP89 TRL#	EOP257	PR Earned Value Target
N/A	N/A	IGR F	Gate F	N/A	N/A	N/A
MRL10	Full Rate Production demonstrated and lean production practices in place	IGR E Production Improvement	Gate E	TRL9 (In service)	Gate 4	100%
MRL9	Low Rate Production demonstrated; Capability in place to begin Full Rate Production	IGR D Serial Production	Gate D	TRL8 (Qualification/Certification) *or TRL9 (In service)*	Gate 3	87.7%
MRL8	Pilot line capability demonstrated; Ready to begin Low Rate Initial Production	IGR C2 Low-Rate Production start (LRP)	Gate C2	TRL7 (First Engine to Test FETT) *or TRL8 (Qualification/Certification)*	Gate 2	65.8%
MRL7	Capability to produce systems, subsystems, or components in a production representative environment	IGR C/IGR C1 Production Readiness	Gate C/C1	TRL7 (First Engine to Test FETT)		43.3%
MRL6	Capability to produce a prototype system or subsystem in a production relevant environment	IGR B End of OPS planning phase & Strategies	Gate B	TRL6: Engine Demo / Prototype	Gate 1	24.1%
MRL5	Capability to produce prototype components in a production relevant environment			TRL5: Rig/core (Expanded design space)		
MRL4	Capability to produce prototype component in a laboratory environment	IGR A Preliminary OPS Planning Phase & Strategies	Gate A	TRL4: Rig test (minimal design space)		14.4%
MRL3	Manufacturing Proof of Concept Developed			TRL3: Proof of concept		
MRL2	Manufacturing Concepts Identified			TRL2: Technology concept		
MRL1	Basic Manufacturing Implications Identified			TRL1: Basic principles (Idea) or Technology Gap (need)		

Table 1 – Relations between MRL, IGR, TRL, , and PR Earned Value target

The mandate of the PR process is to meet the overall production readiness maturity of the program throughout the life of the program at the different IGR and Gates and its ability to start the production phase (Entry Into Production).

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For each of the PR deliverables, these functions

- Industrialization Project Manger for POC deliverables;
- Partnership Industrialization Account Manager for partner POC deliverables;
- and Program Industrialization Manager for program deliverables;

have the overarching responsibility to drive each one of these three activities:

1. Identify PR deliverable gaps as per P&WC form 12284 at the program and at the POC level;
2. Define, plan, analyse and execute mitigation action plans with cross-functional teams to address above PR deliverable gaps;
3. Quantify industrialization risks (impact and likelihood) and prioritize/escalate their mitigation accordingly within the different functions.

Moreover, the industrialization activities are governed with various KPI levels:

- Level 1 (L1) KPIs – all new engine programs together - Responsibility PMO Managers
- Level 2 (L2 KPIs – at individual program level – Responsibility of the Program Industrialization Manager
- Level 3 (L3) KPIs – at individual producer or part/POC level – Responsibility of the Industrialization Project Manager / Partnership Industrialization Account manager

The two main KPIs are the PR earned value% and the MRL%. These two L2 KPIs are measured from the PR Deliverables P&WC form 12284 and are calculated as per below's calculation.

<u>Production Readiness Earned Value% Calculation</u>

$$\frac{Actual: Quantity\ of\ PR\ deliverables\ met\ up\ to\ IGR\ gate\ X}{Plan: Total\ of\ applicable\ PR\ deliverables\ up\ to\ IGR\ gate\ X}\ x\ Earned\ Value\ target\ at\ \text{IGR}\ gate\ x$$

= **PR Earned Value%** for Gate X

PR earned value is calculated at the POC level (L3) as well as the program level (L2)

<u>MRL% Calculation</u>

MRL actual for a given POC is driven by the lowest PR deliverable that is not met (e.g. if the lower PR deliverable not met is a IGR deliverable, therefore the POC meets IGR target (the gate before) so the POC is at MRL8 – see table 1 for relations between IGR and MRL targets).

Then the MRL% is calculated at the program level (L2) as follow:

MRL% at IGR gate x = % of POC meeting the MRL target of IGR gate x

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Additional L2 program-level KPIs are listed in Appendix B, including their applicability for each of the IGR gate as well as calculation method, and source of truth. Due to the varying scope and complexity from one program to another, the L2 KP targets may be program-specific, and such target definition are led by the Program Industrialization Manager, and reviewed with PMT and PMMT.

A notional example of the MRL% and EV% progression and projection is shown in Figure 4 below. It shows historical actuals, current value as well as projection until next IGR gate. This graph refers to the MRL S-Curve and PR Earned Value progression curve.

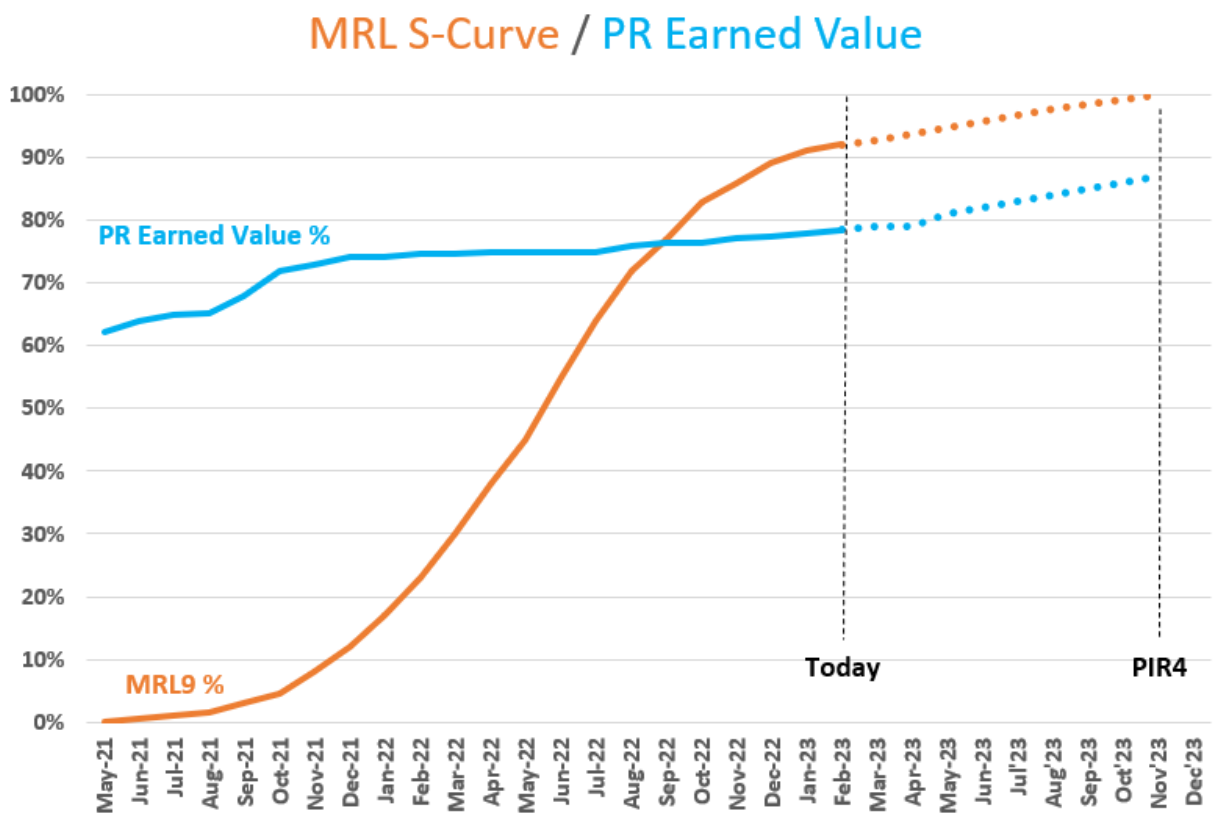

Figure 4 – Notional PR Earned Value progression and MRL% S-Curve– PWC Form 12284

4.3.1.1 MRL Integration within PR

MRL assessment has been integrated as part of the PR deliverables (PWC Form 12284). From the same PR deliverable assessment source, both the MRL% as well as PR Earned Value % are calculated. The PR process is supporting, coordinating and

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centralising the MRL execution across all organizations including: Research & Technologies, Manufacturing Engineering, Program Engineering, Core Engineering, OPO, Supply Management, International Manufacturing, Canadian Manufacturing, Assembly and Test, Partnership office, Quality, and Aftermarket disciplines (refer to Appendix A to get full list of PR deliverables). The MRL methodology is applied on all selected POCs, for all new engine programs at PWC (from small to large derivative and new centerline).

The PR Deliverable Assessment tool (PWC Form 12284) is at the base of the standard work used to assess the progress of all PR deliverables (including the mitigation action plan and associated risks) against the PR deliverables to be achieved for each of the IGR gates. The PR deliverable file content can be seen within the industrialization Power BI where we can create views by POC, producers and/or programs. This assessment tool is populated by the Industrialization Project Manager and Program Industrialization Manager based on various inputs collected with the PR reviews, the IPT meetings and with the feedback of the various business collaborators, i.e. Project Engineers, Research and Technology team, Operations Program Engineers, NPI Buyers, Quality Engineers, etc.

The PR Deliverable Exemption form #12548 could be used to get one or many PR deliverable exempted for one or several POC. The Accountable team, as defined in P&WC form 12284 RACI has the authority to approve this exemption and the approved exemptions must be stored at this location : PR Deliverable Exemption Archive Folder

4.3.2 Industrialization Scope of New Programs and Parts Of Concern (POC) Identification

Before IGR A is completed (Permission to offer), the potential introduction of a new program triggers the determination of the initial industrialization scope. This scope is led by the Program Industrialization Manager with the support of a cross-functional team.

By default, the initial POC selection at IGR A will be all the unique to the program parts as well as the common parts (legacy) having recurrent producibility issues.

The initial POC selection should be provided by the Program Industrialization Manager to OPM and program engineering to ensure that these parts can be included as part of the scope of the program and budgeted accordingly as part of the Operations Start-Up Budget and Engineering Cost Estimate (ECE) budget at Gate A.

Then between IGR A and IGR B, and after the P&WC contract award is confirmed by the customer (“The Program Go"), a revised POC Selection needs to be defined. The scope of this selection is determined by a set of POC criteria (Ref. WIM 693 and WIM 1048) and is reviewed between the Program Industrialization Manager, Manufacturing Engineering as well as Operations Program Management. Thereby, the scope will converge to a reduced list of POCs. This list of POCs becomes the industrialization scope and it will be used for performing the PR deliverable assessment at the program and producer level using form 12284.

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Finally, throughout the rest of the NPI cycle and until Gate F is completed, the Dynamic POC process (Ref. Figure 5) will review the industrialization scope (list of POCs) of all programs at a minimum on a quarterly basis. This review will be governed by the production readiness organization and by getting input from various organization (Supply Management, OPM, Engineering, Aftermarket). This dynamic POC approach starts with the industrialization risk analysis of all PN contained in all NPI programs ensures that no major risk is generated by a part not initially categorized as a POC, e.g. recurring quality history, new supplier introduction. Throughout the dynamic POC process, POC will be added and some would be removed based on their industrialization risk level (PN with more risk are kept as POC).

The list of identified POCs by program, including the Industrialization Project Manager assignments, is maintained in the PR Deliverable assessment tool. This list is maintained by the Program Industrialization Manager and the Industrialization Project Manager. POC can also be added or removed whenever required. In addition, upon the review and update of the POC list, the PR Manager reviews the capacity planning of the PR organization to ensure the capacity level of the organization can support the PR activities for all the selected POCs, and all NPI programs.

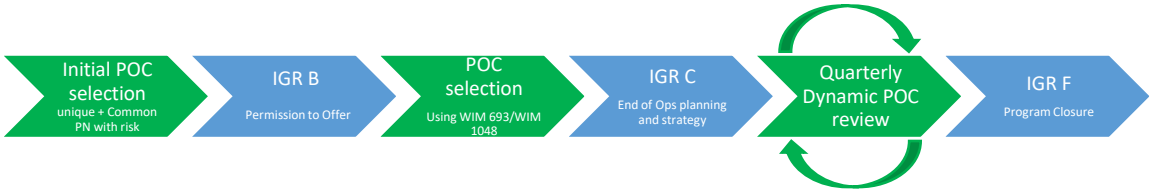

Figure 5 – POC Selection Process

4.3.3 Production Readiness Reviews

4.3.3.1 Producer PR Review

The POC PR Deliverable assessment is done internally by the Industrialization Project Manager/ Partnership Industrialization Account Manager (IPM/PIAM) which are responsible to collect all pertinent data to get an accurate industrialization assessment of its maturity at any given industrialization phase. Once the initial PR deliverable assessment is done, the PR review cycle will start with the support of the various business attendees. The Producer PR reviews allow the IPM/PIAM to understand the applicable PR scope for this specific POC, to get an accurate understanding of the POC maturity and to define the required actions to fulfill the gaps. The applicable PR scope can be all PR deliverables, or a condensed set of deliverables based on a risk-based approach, or a set of deliverables jointly defined by the Program Industrialization Manager and OPM. During the PR reviews, the actions and risks

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are recorded by the IPM/PIAM in form 12284 (PR Deliverable). The IPM/PIAM is responsible to coordinate the PR reviews at a reasonable frequency with the adequate business attendees. The NPI buyer or the Account Specialist are required at these reviews unless otherwise specified differently by them. Refer to table 2 below for frequency of the review, proposed attendees and review expected deliverables.

4.3.3.2 PR-L2

In preparation of the IGR or in case an escalation is needed, the Industrialization Project Manager/Partnership Industrialization Account Manager may plan the PR-L2 (Production Readiness Level 2) review. This meeting is required only for POC considered at high risk of not meeting the Gate exit criteria and for program partners. Director and/or Executive leaders of relevant organizations will participate in this review where industrialization gaps and their detailed mitigation plans will be reviewed. Refer to table 2 below for frequency of the review, proposed attendees and review expected deliverables.

4.3.3.3 Program Level Event

When a program needs a specific event (which is recommended to be a face-to-face meeting/workshop), the Program Industrialization Manager will plan a Program Level Event. This event can be of various nature, e.g. 3P, Business Process Mapping, Value Stream Mapping, D4C, D4A, Industrialization white paper strategy meeting, etc. The Program Industrialization Manager is responsible to communicate the results of this event and to coordinate and track the actions for closure. Refer to table 2 below for frequency of the review, proposed attendees and review expected deliverables.

4.3.3.4 Industrialization Gate Review (IGR)

IGR are used to provide the industrialization program status in preparation of a gate (Gate A to Gate F) and to measure the program progress in between the gates. Gate A and Gate C, i.e. IGR B, and in between the gates Gate C and Gate C2, i.e. IGR C1 (refer to figure 3 for additional details). These reviews are meant to detail the status of the Exit criteria and PR deliverables for the program, identify industrialization risks, as well as their related mitigation action plans. The summary of the IGR results is communicated at the gates. The Program Industrialization Manager is responsible to trigger the preparation and the coordination of the IGR reviews according to the program timeline. Refer to table 2 below for frequency of the review, proposed attendees and review expected deliverables.

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Each IGR review shall contain at a minimum the following elements and a IGR template is documented as part of P&WC form 12532

- IGR Agenda
- Previous IGR open action review
- Program Background
    - Program commercial summary
    - Incremental volumes
    - Engine Concept
- IGR summary
    - Industrialization L1 plan
    - Exit Criteria assessment (go/conditional go/no go)
    - Industrialization or Program Risk Cube
    - Program KPI
- Action plan for Exit criteria gaps
    - For each exit criteria that is not at GO, a detailed mitigation action plan, including action owners and expected completion date will be presented. If mitigation does not meet program requirement an escalation will be identified
- Recommendation and next steps

In table 2 below are the proposed frequency, proposed attendees as well as the expected deliverables out of the various PR process reviews.

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PR Review Type	Proposed Frequency	Proposed Attendees	PR Review deliverables
Producer PR Review	In-line with POC maturity and program milestones: high maturity->low frequency, low maturity->high frequency	IPT Lead, MA/ME, OPE, NPI Buyer, , AS Producer Industrialization Project Manager R&T (if TRL<6)	Production Readiness Dashboard Mitigation Plan update Discrepancy Reports Production Readiness Dashboard
PR-L2	In-line with POC maturity and IGR	PMT Team Engineering Program Manager, OPM Manager IPT Lead, MA/ME, OPE, NPI Buyer, , AS Industrialization Project Manager Program Industrialization Manager Value Stream Management Team GT Professional Producer Executive Team	PR-L2 Executive Review leveraging the P&WC IGR template form# 12532 as a guideline template Exit Criteria (Go – Conditional Go – No Go) Mitigation Plan Review
Program Level Event	Whenever deemed necessary (event funding to be approved by PMMT)	Various – depending the event and the scope	Production Readiness integration, validation and simulation Events and/or workshops Action plan
Industrialization Gate Review	In-line with program gates and milestones, e.g.: Gate, PDR, CDR, EIP, EIS, etc.	O-PMMT typical audience: VP Operations VP Industrial Management VP Supply Management PMO Director OPM Director Marketing Director MA/ME Director Quality Director SM Director Strategic Sourcing Director Engine Center Director CMO Director GM&T Director PR Management Product Cost Management Engineering Program Management GT Professional Industrialization and other designated Fellows	Exit Criteria Status (Go, Conditional Go, No Go) Gap mitigation plan with owner and expected completion dates Industrialization Risk and mitigation plan Production Readiness L2 KPIs

Table 2 – Production Readiness Review Formats

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4.3.3.5 Standard for PR Deliverable & Exit Criteria Reporting

The PR Deliverable Assessment tool uses a binary approach (met/not met) for each applicable PR deliverable. Each of the PR deliverable is contained within one of the 10 PR Exit Criteria:

1. Planning Phase
2. Configuration
3. Product Cost
4. Sourcing
5. OTD Assurance Plan
6. Quality
7. Producibility
8. Assembly & Test
9. Global Chemical Regulations
10. Global Trade

At IGR and Gate reviews, each one of the exit criteria are to be presented with one of the three following decisions (Go/No-Go/Conditional). The recommendations are following this standard as per table 3 below:

Status of the Criterion	Recommendation/Color	Action Required
Completed at the time of the IGR or Gate Review	Go	No action required
Open at the time of the IGR or Gate Review	Conditional Go	Mitigation plan in place for program recovery, small impact on the program
Open at the time of the IGR or Gate Review	No Go	No or partial mitigation plan in place – High impact on the program – Escalation is needed

Table 3 – Standard for Exit Criteria and Dashboard Reporting

For each of the Exit criteria that are not at a Go status, a gap mitigation plan with action owners and commitment dates will have to be presented.

Each of the 10 Exit Criteria are divided into specific PR deliverables as per P&WC form 12284. The maturity of the deliverables is assessed with the PR earned value and MRL% formulas as per section 4.2.1. The PR deliverable are of two types:
- POC type – applicable at the POC level
- Program type – Applicable at the program level (not specific to a POC or vendor)

For each POC/Program deliverables, the Industrialization Project Manager/Program Industrialization Manager/Partnership Industrialization Account Manager are responsible to identify the applicable deliverables based on customer/program and quality requirements, risk assessment and feedback/inputs coming from the various business program collaborators, i.e. GT Focal, GCR Specialist, NPI Buyer, Research and Technology Engineer, Project Engineer, MA/ME, OPE, Quality Specialist, SQAR, etc. For more details on RACI for each one of the deliverable refers to section 5.1.1 below.

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For additional details on the PR Deliverables, Exit criteria and elements, refer to Industrial Management Intranet site: PRM 07-01

4.4 INDUSTRIALIZATION GATE REVIEWS, EXIT CRITERIA AND KEY PROCESS INDICATORS (KPI)

4.4.1 IGR A Objectives

The goal of this phase is to capture customer inputs, benchmark data, lessons learned, regulatory requirements, technical specifications, company know-how, and strategies into a product concept and realization plan. The OPM organization and the Program Industrialization Manager are actively participating to the definition of a white paper (project plan) integrating all the strategies and operational activities of the program. In addition, the OPM organization and the Program Industrialization Manager provide operation requirements as inputs to the PDRS. This phase ends with a satisfactory RLPM gate review (Gate A) and IGR A.

Based upon the results of the IGR A, the vice-president of Operations and/or Industrial Management may need to request a trigger gate when a significant risk is compromising the objectives of the program. Refer to table 4 below for IGR A description and objectives.

IGR Requirements

IGR A requirements include the following:

1. Preliminary Program L1 plan containing:
    a. Product design concept
2. Preliminary Production plan containing:
    a. Industrialization strategy
    b. Producibility risk identification
    c. Operations Budget & staffing
    d. Preliminary sourcing strategy
    e. Establish requirement for partners participation
    f. A Factory Standard Cost (FSC) study (refer to MMP 501) summarized in preliminary Production Plan
3. New Technology planning
4. A global Trade review conducted and baseline Export/Import (EXIM) Plan, if required

4.4.2 IGR B Objectives

The goal of this phase is to capture customer inputs, benchmark data, lessons learned, regulatory requirements, technical specifications, company know-how, and strategies into a product concept and realization plan. The OPM organization and the Program Industrialization Manager are actively participating to the definition of a white paper (project plan) integrating all the strategies and operational activities of the program. In addition, the OPM organization and the Program Industrialization Manager provide operation requirements as inputs to the PDRS.

IGR Requirements

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IGR B requirements include the following:

1. A baseline Business plan (see section 6 of mmp220this document) highlighting
    a. Required phase funding, schedule and resources/investments
    b. Management Reserve identified per P&W WP 1.5 aligned with program risks identified (refer to MMP515 for standard work)
2. A Draft program Plan (see section 6 of this document) highlighting
    a. A defined program scope including customer and internal requirements captured in a preliminary PDRS (see EOP 49)
    b. Funded TRL/MRL maturation plan aligned with requirements in EOP 31 – Engineering Cost estimates (refer to MMP515 for standard work)
    c. Labour requirements from Operations to support program L1 plan, associated risks and critical skills identified.
    d. Level 1 Plan (Creation of the proposed timeline including AR approval)
    e. The intended partners and/or suppliers and the proposed quality plans as well as any additional company requirements
    f. The program archetype and tailoring review/approval for this program
    g. A Partnership/Collaboration strategy, including Partnership assessment (per MMP 507), work scope distribution (per EOP 102), and Program share allocation
3. A comprehensive assessment of the overall program risk associated with the Production plan, including:
    a. All items identified as Red must have a mitigation plan presented (Including status of required resources financial and labour)
    b. Identified low TRL and MRL (<6 items) identified and aligned with draft program plan TRL/MRL maturation plan
4. A draft Production Plan (see section 6 of MMP220)
    a. A review of the Factory Standard Cost (FSC)
    b. A preliminary Industrialisation plan
    c. Production Readiness Dashboard Exit Criteria (refer to IMP 07-01-01)
5. The ‘Enterprise KPI’ Deliverables (refer to Form 11721)
6. Global Trade (GT) assessment status and export licensing plan (Birth Certificate)
7. Draft Development Plan (see section 6 of MMP220)
8. A program communication strategy

4.4.3 IGR C Objectives

The goal of this phase is to translate the PDRS, quality, and cost requirements into a controlled, verified, and validated product design. Design validation is achieved using prototype, development, or production parts in test environments that can represent the customer’s installation and subject the product to extreme conditions required by contract or regulation. Additionally, from an operational perspective, this phase is the beginning of the execution set in the white paper (project planning). This phase ends with a satisfactory IGR C preceding Gate review C.

IGR Requirements

IGR C requirements include the following:

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1. A baseline Business plan (see section 6 of mmp220) highlighting
    a. An agreed set of program goals pertaining to Operations
    b. A status from Operations business units on labour requirements identified in Gate B to support program L1 plan, associated risks and mitigation plans
    c. An agreed L1 plan
    d. A product overview and an agreed architecture concept
    e. An agreed PDRS containing regulatory requirements, key customer requirements and internal requirements pertaining to Operations
    f. Any identified non-compliances to PDRS necessary to meet operations program goals, agreed by PMMT and summarized in program plan
2. An approved capital investment plan and NRE status
3. A baselined Production plan (see section 6 of MMP 220) highlighting
    a. An initial assessment of facility, site and infrastructure plans and requirements
    b. An internal initial production capacity plan and/or factory model with a plan to review maturity during technical reviews, including manufacturing automation plans
    c. Sourcing and Manufacturing Strategy (including PPAP plan with a list of unique part numbers including mitigation for common hardware issues) (refer to QMSP 07-05-02)
    d. An initial Production Readiness Dashboard Exit Criteria
    e. A Factory Standard Cost (FSC) summary and plan
4. A comprehensive assessment of the overall program risk impacting Operations, including:
    a. All items identified as Red must have a mitigation plan presented (Including status of required resources financial and labour)
5. TRL and MRL 6, whichever is greater, If this is not achievable:
    a. There must be a proposed TRL/MRL maturation plan to result in achieving TRL level 6 by Preliminary Design Review (PDR) and an MRL 6 by Critical Design Review (CDR)
    b. Funding of the TRL and MRL maturation plan must occur and include back up plans if unable to achieve the proposed TRL/MRL 6 level. TRL/MRL maturation plan shall be aligned with requirements in EOP 31- Engineering Cost estimates and MMP515 guidance
    c. The back up plans must meet both the Program and Business objectives
6. The ‘Enterprise KPI’ Deliverables (refer to Form 11721)
7. Global ITC assessment status and export licensing given the sourcing plan and scope changes

4.4.3.1 IGR C1 Objectives

IGR C1 will be triggered following a successful completion of a PDR as per MMP 220.

The goal of this phase is to design and develop the production processes needed to produce product that consistently fulfill technical, quality, and cost requirements while operating at the customer demand rate. During this phase, a IGR C1 is performed to validate the status and risks regarding a planned entry into production.

IGR Requirements

IGR C1 requirements include the following:

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1. An updated program Plan (see section 6 of this document) highlighting:
    a. An updated to program goals highlighting any differences since Gate C
    b. An update to L1 plan
    c. A product overview and an architecture concept following PDR
    d. Any changes and compliance to documented regulatory, customer or internal requirements impacting Operations
    e. A status of compliance against PDRS including a PMMT agreed set of non-compliance to PDRS requirements pertaining to Operations
2. An updated Business plan highlighting status of key deliverables and any change in scope impacting Operations
3. An update to NRE status
4. An updated Production plan (see section 6 of MMP 220) highlighting:
    a. An initial assessment of facility, site and infrastructure plans and requirements
    b. An internal initial production capacity plan and/or factory model with a plan to review maturity during technical reviews, including manufacturing automation plans
    c. Sourcing and Manufacturing Strategy (including PPAP plan with a list of unique part numbers including mitigation for common hardware issues) (refer to QMSP 07-05-02)
    d. An initial Production Readiness Dashboard Exit Criteria
    e. A Factory Standard Cost (FSC) summary and plan
5. A comprehensive assessment of the overall program risk impacting Operations, including:
    a. All items identified as Red must have a mitigation plan presented (Including status of required resources financial and labour)
6. TRL 6 for those items identified in Gate C requiring TRL progression. MRL 6 for all components, if this is not achievable:
    a. There must be a proposed TRL/MRL maturation plan to result in achieving the MRL 6 by sub-gate C.2
    b. Funding of the TRL and MRL maturation plan must occur and include back up plans if unable to achieve the proposed TRL/MRL 6 level. TRL/MRL maturation plan shall be aligned with requirements in EOP 31- Engineering Cost estimates and MMP515 guidance
    c. The back up plans must meet both the Program and Business objectives
7. The ‘Enterprise KPI’ Deliverables (refer to Form 11721)
8. Global ITC assessment status and export licensing given the sourcing plan and scope changes

4.4.3.2 IGR C2 Objectives

IGR C2 will be done ahead of Sub-Gate C2 per MMP 220.

The goal of this review is to ensure pre-production or production engine hardware provisioning inventory can be authorized within the guidance defined per MMP 220 while ensuring appropriate program readiness to achieve customer schedule commitments.

IGR Requirements

IGR C2 requirements include the following:

1. An updated program Plan (see section 6 of this document) highlighting:
    a. An update on the program status

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   b. An updated to program goals highlighting any differences since Gate C1
   c. An update to L1 plan
   d. A review of product performance updated based on preliminary verification of the functionality of the product, shortfalls and roadmap to meet KPIs
   e. An update to compliance status against the PDRS (refer to EOP 49)
2. An updated Business plan highlighting status of key deliverables and any change in scope impacting Operations
3. An update to NRE status
4. An updated Production plan (see section 6 of MMP 220) highlighting:
   a. An update to Sourcing and Manufacturing Strategy (including PPAP plan) including the ramp up to manufacturing plan to support EIP/EIS
   b. A review of the supply chain preparatory work as outlined in ASQE-01
   c. An update on the assessment of facility, site and infrastructure plans and requirements
   d. Sourcing and Manufacturing Strategy (including PPAP plan with a list of unique part numbers including mitigation for common hardware issues) (refer to QMSP 07-05-02)
   e. An update to the Production Readiness Dashboard Exit Criteria
   f. A Factory Standard Cost (FSC) summary and plan
5. A comprehensive assessment of the overall program risk impacting Operations, including:
   a. Any risk of ordering production hardware prior to completion of validation
   b. Any retrofit risks including proposed mitigation plans
   c. All items identified as Red must have a mitigation plan presented (Including status of required resources financial and labour)
6. TRL and MRL > 6 of the product. If this is not achievable:
   a. There must be a proposed TRL/MRL maturation plan or back up plan with funding secured to result in achieving the MRL > 6 meeting Program and Business objectives
7. Conversion of the engineering BOM (eBOM) into manufacturing BOM (mBOM) or agreed PMMT plan for change control
8. The 'Enterprise KPI' Deliverables (refer to Form 11721)
9. Global ITC assessment status and export licensing given the sourcing plan and scope changes

4.4.4 IGR D Objectives

The goal of this phase is to validate that product fulfills the design requirements and the production processes have demonstrated the capability to consistently produce conforming product at the customer demand rate. Product validation is achieved using product manufactured from the defined production processes. IGR D is to be held ahead of MMP 220 Gate D. The IGR D can be conducted in conjunction with the Pre-Production Readiness Reviews stipulated per MMP 220 within the production phase of the program if first pre-production is more than 3 months before Manufacturing Approval.

IGR Requirements

IGR D requirements include the following:

1. An updated program Plan (see section 6 of this document) highlighting:
   a. An update on the program status
   b. An updated to program goals highlighting any differences since Gate C2
   c. An update to L1 plan

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   d. A review of product performance updated based on preliminary verification of the functionality of the product, shortfalls and roadmap to meet KPIs
   e. An update to compliance status against the PDRS (refer to EOP 49)
   f. Identification of funded program improvement plans based on customer/airframe Market Feedback Analysis, a Customer Oriented Results & Excellence (CORE) method, as required
   g. A Product performance and architecture stability overview (including proof of manufacturing with lessons learned incorporated into design or manufacturing documentation)
   h. The release both the Technical Data Package (TDP - including definition, ESA and SFI approvals) and Manufacturing Data Package (MDP - including MOS, AFS and approved repairs).
2. Conversion of the engineering BOM (eBOM) into manufacturing BOM (mBOM) while highlighting pending configuration changes and production impact.
3. An updated development plan highlighting:
   a. Completion of all qualification testing and completion of all certification reports with remaining open items identified
   b. Completion of all verification activities for first production hardware (reference EOP 57) including PFMEA, DFMEA, FAIR.
   c. Status of compliance against certification requirements
4. An updated Business plan highlighting status of key deliverables and any change in scope impacting Operations
5. An updated Production plan (see section 6 of MMP 220) highlighting:
   a. Released production BOM and production sources identified for all parts with approved rate plans
   b. Gap assessment results of facility, site and infrastructure plans and requirements in support of serialized production
   c. A Factory Standard Cost (FSC) summary and plan established and funded
   d. An update to the Production Readiness Dashboard Exit Criteria
   e. Industrial Strategy (including PPAP plan status as per QMSP 07-05-02) with resources in place (to include but not limited to staffing, equipment, tooling and fixtures)
   f. Documentation of industrial risks and technical debt at gate D with funded mitigation plans in place and communication to the customer as required
   g. A list of identified post certification changes at gate D that require customer or field retrofit (OEUP) and consequential financial impact to business plan assumptions
6. Verification that all DRs having Hazardous Engine Condition or EH&S implications are closed with corrective action (as per EOP 158) and evidence that all other DR have action plans
7. A comprehensive assessment of the overall program risk impacting Operations, including:
   a. A summary of the Industrialization and technical risk identified at the technical review that was held prior to Gate D
   b. Any retrofit risks including proposed mitigation plans
   c. All items identified as Red must have a mitigation plan presented (Including status of required resources financial and labour)
8. TRL 8 and MRL 9 of the product. If this is not achievable:
   a. There must be a proposed TRL/MRL maturation plan or back up plan with funding secured to result in achieving TRL 8 and MRL 9 meeting Program and Business objectives
9. Global Trade assessment status and export licensing plan (Birth Certificate) given the sourcing plan and scope changes

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4.4.5 IGR E Objectives

The goal of this phase is to validate that production is operating at a efficient rate to achieve customer demands while highlighting critical operational flow disruptors. IGR E is to be held ahead of MMP 220 Gate E. The IGR E can be conducted in conjunction with the Production Performance Reviews stipulated per MMP 220 within the production phase of the program.

IGR Requirements

IGR E requirements include the following:

1. An updated program Plan (see section 6 of this document) highlighting:
    a. An update on the program status
    b. An update to L1 plan
    c. An update to compliance status against the PDRS (refer to EOP 49)
    d. Identification of funded program improvement plans based on customer/airframe Market Feedback Analysis, a Customer Oriented Results & Excellence (CORE) method, as required
2. An updated Business Plan (see section 6 of MMP 220) highlighting performance against contractual obligation assessment to determine customer and business case impact
3. A comprehensive assessment of the overall program risk impacting Operations, including:
    a. A summary of the Industrialization and technical risk identified at the technical review that was held prior to Gate review (refer to EOP 103 & PDD0250)
    b. Any retrofit risks including proposed mitigation plans
    c. All items identified as Red must have a mitigation plan presented and associated post certification investment financial resources
4. Capture and communicate operations lessons learned for future programs
5. An updated Production plan (see section 6 of MMP 220) highlighting:
    a. An update to the manufacturing resources (such as staff and equipment) plan
    b. An update on Factory Standard Cost (FSC)
    c. If applicable a plan for all government or customer furnished/owned property
    d. If applicable, completion of all production contractual requirements
    e. An update to the Production Readiness Dashboard Exit Criteria
    f. Completion of MRL assessments (demonstrate that product is MRL 9, show MRL readiness plan for new product changes)
    g. Health on engine and component spares delivery to support the aftermarket
    h. If applicable, a preliminary end-of-Production plan including a plan for manufacturing resources and asset/property transition
6. An update to the defined Enterprise KPI deliverables (form 11721)
7. NRE status on the program
8. Global Trade review conducted and updated the export/import plan as required
9. If applicable, a communication plan including discontinuation notice for internal and external impacted parties

4.4.6 IGR F Objectives

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The goal of this review is to solicit agreement and alignment to discontinue support of a product or service and is to be conducted as required ahead of MMP 220 Gate F.

IGR Requirements

IGR F requirements include the following:

1. Status on completion of contractual obligations and customer impacts review
2. Completion of discontinuation production strategy (to include plan for manufacturing resources, all government or customer furnished/owned property, the cost/benefit analysis for ending production, end-of-Production plan, including resource plan and asset/property transition)
3. A communication plan including discontinuation notice for internal and external impacted parties
4. Approved risk management plans, if necessary
5. Consensus on program exit strategy
6. Consensus on trigger condition points through PMT disbanding

4.4.7 Exit Criteria and Key Process Indicators (KPI)

The exit criteria are the conditions that have to be fulfilled in order for a IGR review to be considered complete. PR deliverables included for each of the IGR varies and they are aligned with the scope of the program and its development cycle. To consult all criteria and elements for each phase, refer to appendix A.

Note: see Appendix B for the list of PR KPIs.

4.5 PR TRANSFER TO LEGACY BUSINESS PROCESS

4.5.1 POC Transfer

After the completion of the IGR D, the Program Industrialization Manager starts to review the list of POC to assess if the PR process can be closed on those. To conclude the PR activities on a given POC, the transfer of the following 4 functions must be completed as per transfer triggers in Table 9.

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Function	Before transfer	After transfer	Triggers
Supply Management	NPI SM	Account Specialist (AS)	PPAP interim B + other requirements listed as part of PWC Form 10865E (NPI to AS Transfer Checklist)
Producibility / Quality	MA/ME/Partnership Quality	QS	This transfer happens when the full approval or any exception preventing Interim A to go to full (as per QMSP 07-05-02)
Engineering	Program PE	STO PE	First to occur between: - Gate D - 2 years after EIS
Industrialization	PIM/IPM/PIAM	Operations Program Management	MRL10 at IGR E + No active retrofit campaign on pre-production engines

Table 9 – POC activity transfer

The conclusion of the PR activities is driven by the Program Industrialization Manager during a review conducted with the following stakeholders listed in table 9 above. The transfer is recorded in the PR deliverable assessment. The Industrialization Project Manager must follow-up after the transfer period until all actions are closed and/or action ownership has been agreed upon.

The complete transfer for the 4 functions above has to be fully completed and agreed by the giving and receiving parties before the IGR E.

4.5.2 Internal Capacity for Operations

Direct resources required to execute on the new engine program deliverables and milestones are accounted as part of the ECE and Operations Start-Up budget and respectively managed by the program engineering and OPM team.

For indirect resources, their availability to support the new engine program are evaluated by the Program Industrialization Manager. He is also responsible to communicate to all the involved functions what they will have to consider in their resource planning to support the new engine program needs.

At each IGR the Program Industrialization Manager will review with but not limited to: SM, Quality, GM&T, Regional Office, and PMO functions their internal capacity resource planning and the actions required for supporting the program needs until IGR E.

More specifically to SM, the Program Industrialization Manager will provide them the forecast of all the new parts and new suppliers to be handled by the material planning organization.

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SM organization will need forecast on quantity and introduction timeline for unique parts, new suppliers or new supplier's location, as well as program specific conditions or systems that could impact existing processes.

The Program Industrialization Manager will provide to SMQ all the new suppliers to be managed with the NSD process according to the program milestones.

5 ORGANIZATION & RESPONSIBILITIES

5.1.1 Industrial Management – Organizational Chart

The figure 6 shows the functional organization of the Operations Program Office group.

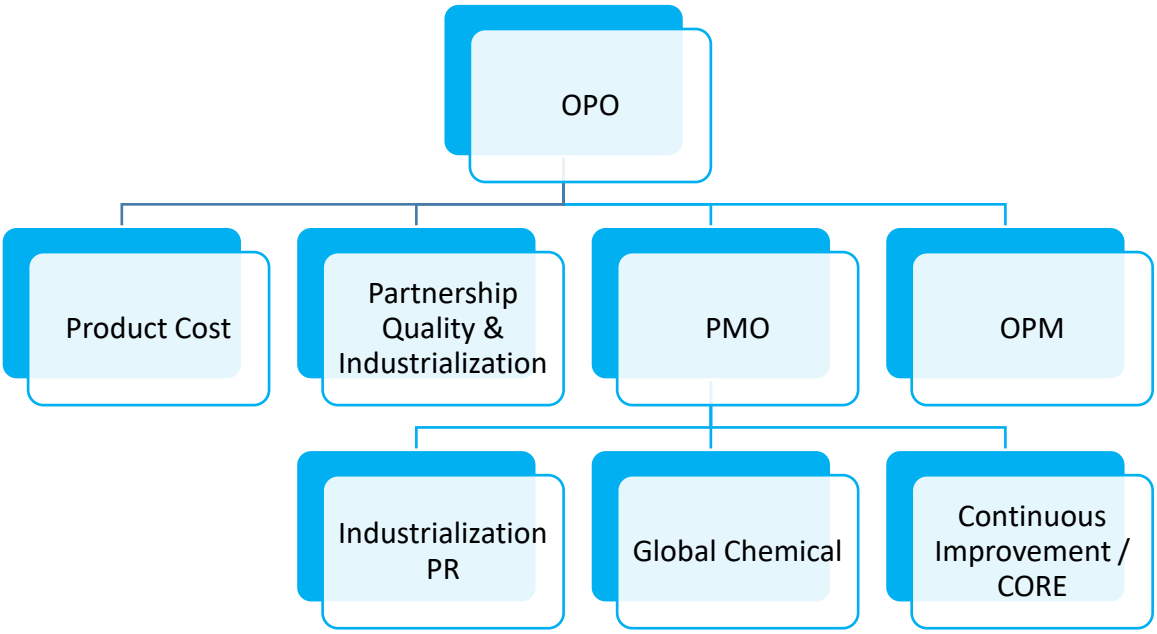

Figure 6 – Industrial Management Functions - Organizational Chart

5.1.2 RACI Chart

The following RACI chart shows the distribution of the tasks by responsibility levels (R-A-C-I) for each of the PR deliverables. The various functions are not limited only to these responsibilities.

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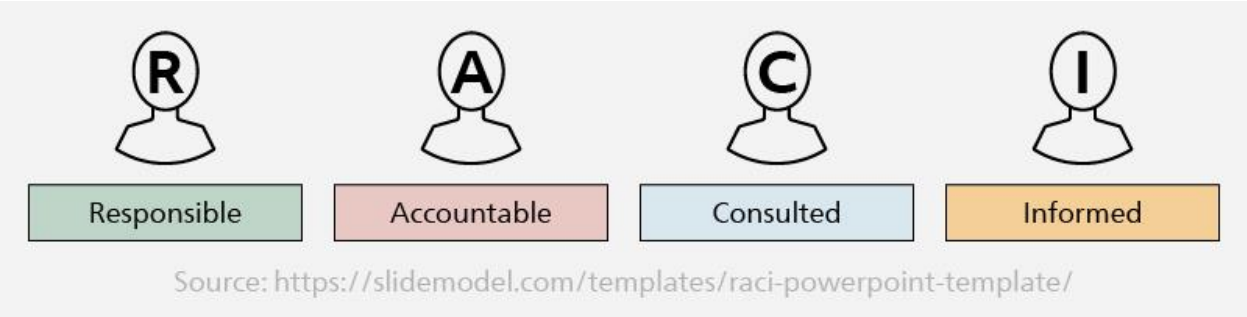
A **RACI** matrix is also known as a responsibility assignment matrix. It describes the participation by various roles in completing tasks or deliverables for a project or business process.

- **R = Responsible:** Those assigned to complete the task/deliverable.
- **A = Accountable:** Those ultimately answerable for the correct and thorough completion of the task or deliverable. This person owns the task/deliverable and has final approval authority. There should be only one person accountable per each task or deliverable.
- **C = Consulted:** Often, subject matter experts sought out for opinions and input (two-way communication)
- **I = Informed:** Those affected by the outcome of a task or deliverable, who are kept up to date on progress, typically when a task or deliverable is complete (one-way communication).



In case of discrepancy between this list below and the one in the P&WC form 12284, the later one would prevail. As P&WC form 12284 is expected to be updated more frequently than the IMP 07-01-01 document, this table below could be only used as general guidance.

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1- Planning Phase							
PR Element	PR Deliverable	Program	POC	Responsible	Accountable	Consulted	Informed
1.1 Industrialization white paper	1.1.1 Industrialization White Paper generated, approved and updated as need be	X		Lead: OPM Co-Lead: Program Industrialization Manager	OPM	Process Owner: Industrialization PR Product Cost Marketing Program PE MA/ME Strategic Sourcing NPI SM/CMO/GM&T Assembly & Test Quality General Counsel Aftermarket Partnership Quality & Industrialization R&T	OPMT PMT PMMT O-PMMT
1.2 Industrialization Planning	1.2.1 Industrialization L1 plan is available and meets program timeline	X		Lead: OPM Co-Lead: Program Industrialization Manager	OPM	Process Owner: PMO+OPM Marketing Program PE Industrialization Fellows Partnership Quality & Industrialization	OPMT PMT PMMT O-PMMT
1.3 Budget	1.3.1 Operations start up budget is supporting program needs	X		Lead: OPM PR/PPAP cost estimation: Program Industrialization Manager Dual sourcing plan: Strategic Sourcing, OPM, Operations Strategy Indirect resource planning: Program Industrialization Manager	OPM	Process Owner: OPM MA/ME NPI SM/CMO/GM&T Strategic Sourcing Marketing	OPMT PMT PMMT O-PMMT
1.4 Industrialization Scope	1.4.1 Program Industrialization KPI targets are defined and revised if needed at IGR gates	X		Lead: Program Industrialization Manager	Program Industrialization Manager	Process Owner: PMO+OPM O-PMT Partnership Industrialization Account Manager OPM	PMT
1.4 Industrialization Scope	1.4.2 Part Of Concerns (POC) are identified and Industrialization Project Manager assigned	X		Lead: Program Industrialization Manager Co-Lead: MA/ME & Industrialization Project Manager	Program Industrialization Manager	Process Owner: PMO+OPM OPM NPI SM/CMO/GM&T R&T (TRL <6 parts)	O-PMT
1.4 Industrialization Scope	1.4 3 PR transfer plan including transfer triggers are clearly defined and agreed with "transferring from" and "transferring to" organizations. Program is meeting NPI to AS% and MA to QS% IGR target.	X		A) Supply Management Lead: NPI SM B) Producibility/ Quality Lead: MA/ME/Partnership Quality C) Industrialization Lead: Program Industrialization Manager/Industrialization Project Manager/Partnership Industrialization Account Manager D) Engineering Lead: Program PE	Program Industrialization Manager	Process Owner: Industrialization PR A) Supply Management: AS B) Quality: QS C) Industrialization: OPM D) Engineering: STO PE OPM	OPMT PMT PMMT O-PMMT
1.4 Industrialization Scope	1.4.4 PR transfer plan between the "transferring from" and "transferring to" organizations is completed. Program is meeting NPI to AS% and MA to QS% IGR target.	X		A) Supply Management leads: NPI SM and AS B) Producibility/ Quality leads: MA/ME/PQ and QS C) Industrialization Leads: Program Industrialization Manager/Industrialization Project Manager/Partnership Industrialization Account Manager / OPM D) Engineering Leads: Program PE and STO PE	Program Industrialization Manager	Process Owner: Industrialization PR PMT OPM	OPMT PMT PMMT O-PMMT

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2- Design Maturity							
PR Element	PR Deliverable	Program	POC	Responsible	Accountable	Consulted	Informed
2.1 Design Maturity	2.1.1 The preferred product concept is established	X		Lead: Program PE Operations PDRS Lead : OPM	Engineering Program Manager	Process Owner: Engineering Marketing Product Cost CMO / GM&T A&T Aftermarket	O-PMT
2.1 Design Maturity	2.1.2 Pre-detail design phase (PDR) completed		X	Lead: Program PE Operations PDRS Lead: OPE	Engineering Program Manager	Process Owner: Engineering Product Cost IPT	O-PMT
2.1 Design Maturity	2.1.3 Formal Concept Exit Review (CER) is completed	X		Lead: Program PE Operations PDRS Lead: OPM	Engineering Program Manager	Process Owner: Engineering Product Cost IPT	O-PMT
2.1 Design Maturity	2.1.4 CDR is completed		X	Lead: Program PE Operations PDRS Lead: OPE	Engineering Program Manager	Process Owner: Engineering Product Cost IPT	O-PMT
2.1 Design Maturity	2.1.5 Engineering BOM (EBOM) is released	X		Lead: Program PE Operations MBOM Lead: OPM	Engineering Program Manager	Process Owner: Engineering Product Cost IPT	O-PMT
2.1 Design Maturity	2.1.6 ESA package status plans are in progress to meet first production part delivery date and be at approved status at Engine certification		X	Lead: Program PE Operations Lead: OPE	Engineering Program Manager	Process Owner: Engineering IPT NPI SM (for buy Parts) CMO/GM&T	O-PMT Partnership Quality
2.1 Design Maturity	2.1.7 ESA% program target is met	X		Lead: Program PE Operations Lead: OPE	Program Industrialization Manager	Process Owner: Engineering IPT NPI SM (for buy Parts) CMO/GM&T	O-PMT Partnership Quality
2.1 Design Maturity	2.1.8 Drawing status is advanced released in time to support the 1st production engine configuration		X	Lead: Program PE Operations Lead: OPE	Engineering Program Manager	Process Owner: Engineering IPT	O-PMT
2.1 Design Maturity	2.1.9 Configuration stability % program target is met	X		Lead: Program PE Operations Lead: OPE	Lead: Engineering Program Manager Configuration Stability % KPI: Program Industrialization Manager	Process Owner: Engineering IPT	O-PMT
2.1 Design Maturity	2.1.10 All Class 1 CO completed in PLM and implemented for the MA engine		X	Class 1 CO completed in PLM: Program PE SAP implementation: OPE	Engineering Program Manager	Process Owner: Engineering IPT	O-PMT
2.1 Design Maturity	2.1.11 Final Design Review is completed		X	Lead: Program PE	Engineering Program Manager	Process Owner: Engineering Product Cost IPT OPM	O-PMT
2.1 Container Requirements	2.1.12 Dedicated Module Container requirements defined and agreed with the Partner		X	Lead: Program PE Co-Lead: Partnership Industrialization Account Manager	Engineering Program Manager	Process Owner: Engineering OPM IPT Partnership Quality	O-PMT
2.2 Demand planning	2.2.1 Production BOM and engine demand is loaded in SAP	X		Lead: OPM	OPM	Process Owner: OPM PMT	OPMT PMT PMMT O-PMMT
2.3 TRL	2.3.1 No technologies below TRL4 are on-boarded	X		Lead: R&T Co-Lead: Program PE	Engineering Program Manager	Process Owner: Engineering IPT OPM	O-PMT
2.3 TRL	2.3.2 Sub TRL-6 technologies on-boarded for the program are identified and technology development plans are supporting program timeline		x	Lead: R&T Co-Lead: Program PE	Engineering Program Manager	Process Owner: Engineering IPT OPM	O-PMT
2.3 TRL	2.3.3 New technologies at TRL 6		X	Lead: R&T Co-Lead: Program PE	Engineering Program Manager	Process Owner: Engineering IPT OPM	O-PMT
2.3 TRL	2.3.4 New technologies at TRL 7		X	Lead: Program PE Co-Lead: R&T	Engineering Program Manager	Process Owner: Engineering IPT OPM	O-PMT

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2.3 TRL	2.3.5 New technologies at TRL 8		X	Lead: Program PE Co-Lead: R&T	Engineering Program Manager	Process Owner: Engineering IPT OPM	O-PMT
2.4 Electronics & Material Obsolescence	2.4.1 Parts with LTB or BB have a mitigation plan to support life of the engine program	X		Lead: Technical Project and Obsolescence Management Operations PDRS Lead: OPM	VSM	Process Owners: Supply Management & Engineering NPI SM Program PE / STO PE OPM	O-PMT
3- Product Cost							
PR Element	**PR Deliverable**	**Program**	**POC**	**Responsible**	**Accountable**	**Consulted**	**Informed**
3.1 Product Cost Status	3.1.1 CBN is established and approved	X		Lead: Product Cost Co-Lead: OPM & Engineering Program Manager	Engineering Program Manager	Process Owner: Product Cost CMO/GM&T MA/ME Project Engineering Aftermarket Marketing VSM NPI SM Strategic Sourcing SFV Partnership office A&T	PMT
3.1 Product Cost Status	3.1.2 FSC projection at learned out year is meeting Target cost baseline	X		Lead: Product Cost Co-Lead: OPM & Engineering Program Manager	Program Engineering: For design related cost drivers (up to and including IGR C2) OPM: Remaining aspects of cost drivers (full accountability after IGR C2)	Process Owner: Product Cost CMO/GM&T MA/ME Project Engineering Aftermarket Marketing VSM NPI SM Strategic Sourcing SFV Partnership office A&T	PMT
4- Sourcing							
PR Element	**PR Deliverable**	**Program**	**POC**	**Responsible**	**Accountable**	**Consulted**	**Informed**
4.1 Production Sourced	4.1.1 Make/Buy/partnership strategy is confirmed and preliminary sourcing strategy is defined on unique and common PN	X		Buy Lead: NPI SM Make Lead: CMO/GM&T Partnership Lead: Partnership Office	Make/Buy: Operations Strategy Partnership: Partnership Office P-SSRB: OPM	Process owner: Strategic Sourcing OPM Partnership Quality & Industrialization VSM Product Cost	O-PMT
4.1 Production Sourced	4.1.2 Confirm whether development source is the same as production source		X	Lead: NPI SM/CMO/GM&T Co-Lead: OPE	OPM	Process Owner: Supply Management OPM MA/ME Product Cost Program PE Strategic sourcing	PMT
4.1 Production Sourced	4.1.3 LTA is timely signed with the final production source to secure program HW requirement		X	Lead: NPI SM Co-Lead: Strategic Sourcing Contracts: Legal (SM Contracts) Partners: Partnership office	OPM	Process Owner: Supply Management Program PE	O-PMT
4.1 Production Sourced	4.1.4 Program LTA% target is met	x		Lead: NPI SM Co-Lead: Strategic Sourcing Contracts: Legal (SM Contracts) Partners: Partnership office	Lead: OPM LTA % KPI: Program Industrialization Manager	Process Owner: Supply Management Program PE	O-PMT
4.2 Sub-tier Management	4.2.1 Producer has provided sub-tier management evidence related to supply security, performance and quality		X	Lead: NPI SM/CMO/GM&T/Partnership Industrialization Account Manager Co-Lead: SQAR/Industrialization Project Manager	NPI SM/CMO/GM&T/ Partnership Industrialization Account Manager	Process Owner: Supply Management Legal (SM Contracts) Partnership Quality SDL	O-PMT
4.3 Dual Sourcing	4.3.1 P&WC Operations and/or program funding is supporting dual sourcing activities to be completed before EIS		X	Lead: NPI SM/CMO/GM&T Co-Lead: Program PE NPI Startup Budget: OPM Operations Managed budget: Strategic Sourcing	OPM	Process Owner: Strategic Sourcing Engineering Program Manager MA/ME Partnership Quality & Industrialization SDL SM Transition	O-PMT
4.4 Orderable details sourcing	4.4.1 Orderable details sourcing is secured	X		Lead: OPM	Customer Service	Process Owner: Customer Service Spares Planning	

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						NPI SM/AS/Strategic sourcing CMO/GM&T	
4.4 Orderable details sourcing	4.4.2 Spare parts demand is loaded in SAP		X	Lead: OPM Planner Co-Lead: Partnership Industrialization Account Manager	Customer Service	Process Owner: Customer Service Partnership Office OPM	O-PMT
5- OTD Assurance Plan							
PR Element	**PR Deliverable**	**Program**	**POC**	**Responsible**	**Accountable**	**Consulted**	**Informed**
5.1 On-Time Delivery	5.1.1 Production hardware procurement plan is supporting first production engine hardware store date.		X	Lead: OPE Co-Lead: NPI SM / CMO / GM&T	OPM	Process Owner: OPM Program PE	O-PMT
5.1 On-Time Delivery	5.1.2 Producer is compliant to specific OTD assurance clauses in LTA, i.e. B2B, 4WBA, OTD if applicable.		X	Lead: NPI SM or AS / CMO / GM&T	Lead: NPI SM or AS / CMO / GM&T	Process Owner: Supply Management VSM OPM PIAM	O-PMT OPM
5.1 On-Time Delivery	5.1.3 Program Build Ahead% target is met	X		Lead: NPI SM or AS / CMO / GM&T	Lead: NPI SM or AS / CMO / GM&T Build Ahead % KPI: Program Industrialization Manager	Process Owner: Supply Management VSM OPM PIAM	O-PMT OPM
5.1 On-Time Delivery	5.1.4 POC is meeting MRP dates for the progam (95%+ OTD at the PN) or has a robust plan to get to MRP by IGR E		X	Lead: NPI SM or AS / CMO / GM&T	Lead: NPI SM or AS / CMO / GM&T	Process Owner: Supply Management VSM OPM PIAM	O-PMT Partnership office
5.1 On-Time Delivery	5.1.5 Program HW OTD% target is met	X		Lead: NPI SM or AS / CMO / GM&T	Lead: NPI SM or AS / CMO / GM&T HW OTD % KPI: Program Industrialization Manager	Process Owner: Supply Management VSM OPM PIAM	O-PMT Partnership office
5.2 Producer Capacity	5.2.1 Producer has a concrete a 5M capactity plan to support program production demand for ramp up		X	Lead: NPI SM or AS/CMO/GM&T Co-Lead: SDL Partners Lead: Partnership Industrialization Account Manager	Lead Up to IGR D: Program Industrialization Manager Lead After IGR D: OPM Capacity Secured % KPI: Program Industrialization Manager	Process Owner: Supply Management VSM OPM	O-PMT
5.3 Single Point Of Failure	5.3.1 SPOF identified at Producer and sub-tiers level with mitigation plan in place		X	Lead: NPI SM or AS/CMO/GM&T Co-Lead: SDL Partners Lead: Partnership Industrialization Account Manager Deliverable Assessment, Action plan, Risk mitigation: Industrialization Project Manager/Partnership Industrialization Account Manager	NPI SM or AS/CMO/GM&T	Process Owner: Supply Management MA/ME VSM OPM SQAR Partnership Quality	O-PMT
5.3 Single Point Of Failure	5.3.2 SPOF mitigation plan completed and accepted for production		X	Lead: NPI SM or AS/CMO/GM&T Co-Lead: SDL Partners Lead: Partnership Industrialization Account Manager Deliverable Assessment, Action plan, Risk mitigation: Industrialization Project Manager/Partnership Industrialization Account Manager	NPI SM or AS/CMO/GM&T	Process Owner: Supply Management MA/ME VSM OPM SQAR Partnership Quality	O-PMT
6- Quality							
PR Element	**PR Deliverable**	**Program**	**POC**	**Responsible**	**Accountable**	**Consulted**	**Informed**
6.1 Quality System Compliance	6.1.1 CQP (Customer Quality Plan) or MP (Management Plan)		X	Lead: Partnership Quality Co-Lead: SQAR & Partnership Industrialization Account Manager	QS&P	Process Owner: Quality SM Quality, Project Engineering	PMT OPM

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	initiated, released and reviewed for Gates					OPM Partnership Office	
6.1 Quality System Compliance	6.1.2 Supplier is PMTS approved		X	Lead: SQAR Partners Co-lead: Partnership Quality	SQA MFI	Process Owner: Quality NPI SM OPM	PMT SQAR
6.1 Quality System Compliance	6.1.3 Partner serialized module and sub-components meeting program requirement		X	Lead: Partnership quality Co-Lead: SQA MFI & Partnership Industrialization Account Manager	Program PE	Process Owner: Engineering Partnership Quality & Industrialization OPM	PMT OPM
6.2 New Supplier Development	6.2.1 New Supplier Development (NSD) plan is in place to meet program needs (first production hardware PO can be placed in time)		X	Lead: SQAR Co-Lead: NPI SM/CMO/GM&T Partners: Partnership Quality	NPI SM or AS/CMO/GM&T	Process Owner: SM SM Quality CMO/GM&T Quality OPM Quality Engineering	PMT OPM
6.2 New Supplier Development	6.2.2 NSD is completed (NSE, NSI & NSA), Production PO can be placed		X	Lead: SQAR Co-Lead: NPI SM/CMO/GM&T Partners: Partnership Quality	NPI SM or AS/CMO/GM&T	Process Owner: SM SM Quality CMO/GM&T Quality OPM Quality Engineering	PMT OPM
6.3 Procert	6.3.1 Cpk value ≥ 1.0 on 80% of KPCds (cat 1 & 2) or approved under Exception		X	Lead: MA/ME Co-Lead: NPI SM/CMO/GM&T Approve exception: SM Quality/OPE	Lead: MA/ME	Process Owner: Quality Program PE PPAP Central OPM	O-PMT
6.3 Procert	6.3.2 Cpk value ≥ 1.33 on 80% of KPCds (cat 1 & 2) or approved under Exception		X	Lead: MA/ME Co-Lead: NPI SM/CMO/GM&T Approve exception: SM Quality/OPE	Lead: MA/ME	Process Owner: Quality Program PE PPAP Central OPM	O-PMT
6.3 Procert	6.3.3 Procert % KPI program target is met	X		Lead: MA/ME Co-Lead: NPI SM/CMO/GM&T Approve exception: SM Quality/OPE	Program Industrialization Manager	Process Owner: Quality Program PE PPAP Central OPM	O-PMT
7- Producibility							
PR Element	**PR Deliverable**	**Program**	**POC**	**Responsible**	**Accountable**	**Consulted**	**Informed**
7.1 Producibility Readiness Level (PRL)	7.1.1 For Built to Print only: PRL is initiated and if any gaps action plan is in place		X	Lead: MA/ME	MA/ME	Process Owner: Engineering IPT OPM	O-PMT
7.1 Producibility Readiness Level (PRL)	7.1.2 For Built to Print only: PRL is completed and score > 80% or PSM required at drawing approval		X	Lead: MA/ME	MA/ME	Process Owner: Engineering IPT OPM	O-PMT
7.2 Process Robustness Index (PRI)	7.2.1 PRI self-assessment request sent to selected production source		X	Lead: MA/ME	MA/ME	Process Owner: Engineering IPT OPM	O-PMT
7.2 Process Robustness Index (PRI)	7.2.2 Producer qualified hard Tooling plan is in place to support the first production HW delivery		X	Lead: NPI SM Co-Lead: MA/ME/OPE MB-DMI/TDP/Script: Quality Team CTI file generation: Engineering drafting team	OPM	Process Owner: Engineering Program PE IPT SM Tooling	OPMT PMT PMMT O-PMMT
7.2 Process Robustness Index (PRI)	7.2.3 PRI self-assessment reviewed, producibility risk identified, action plan to close the gaps defined		X	Lead: MA/ME	MA/ME	Process Owner: Engineering IPT OPM	OPMT PMT PMMT O-PMMT
7.2 Process Robustness Index (PRI)	7.2.4 PRI approved and signed by MA		X	Lead: MA/ME	MA/ME	Process Owner: Engineering IPT OPM	O-PMT
7.3 Component Non-Conformance Status	7.3.1 Known producibility risks based on historical data (common or similar p/n) are identified and improvement plan are defined	X		Lead: MA/ME	Program Industrialization Manager	Process Owner: Industrialization PR OPM CMO/GM&T Partnership Office Partnership Quality & Industrialization	OPMT PMT PMMT O-PMMT

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7.3 Component Non-Conformance Status	7.3.2 The sum of producibility improvement action plans, including BS QN, is projecting to meet QNLI/engine targets by engine certification (IGR D) and EIS (IGR E)	X		External Supplier Leads: MA/ME Partners Leads: Partnership Quality CMO/GM&T Leads: Process Planners	Program Industrialization Manager	Process Owner: Industrialization PR OPM CMO/GM&T Partnership Office Partnership Quality & Industrialization	OPMT PMT PMMT O-PMMT
7.3 Component Non-Conformance Status	7.3.3 Producibility improvement action plans is projecting for the POC to reach < 0.1 QNLI/parts by IGR D		X	External Supplier Leads: MA/ME Partners Leads: Partnership Quality CMO/GM&T Leads: Process Planners	Producers: Industrialization Project Manager Partners: Partnership Industrialization Account Manager	Process Owner: Industrialization PR OPM CMO/GM&T Partnership Office Partnership Quality & Industrialization	OPMT PMT PMMT O-PMMT
7.3 Component Non-Conformance Status	7.3.4 < 0.1 QNLI/part received		X	External Supplier Leads: MA/ME Partners Leads: Partnership Quality CMO/GM&T Leads: Process Planners	Producers: Industrialization Project Manager Partners: Partnership Industrialization Account Manager	Process Owner: Industrialization PR OPM CMO/GM&T Partnership Office Partnership Quality & Industrialization	OPMT PMT PMMT O-PMMT
7.4 PPAP	7.4.1 Producer submitted PPAP plan		X	Lead: NPI SM/CMO/GM&T PPAP Applicability: MFP IDS communication: MA/ME PPAP Producer proficiency: PPAP Central Part Family Strategy: MFP & MA/ME	Producers: Industrialization Project Manager Partners: Partnership Industrialization Account Manager	Process Owner: PPAP Central IPT OPM	O-PMT O-PMMT
7.4 PPAP	7.4.2 Interim B is achieved for the 1st production part		X	Lead: NPI SM/CMO/GM&T 11 PPAP elements: 1) Design Record: MFP 2) Design Risk Analysis: MA/ME/Partnership Quality 3) PFD: MA/ME/Partnership Quality 4) PFMEA: MA/ME/Partnership Quality 5) Control Plan: MA/ME/Partnership Quality 6) MSA: MA/ME/Partnership Quality 7) Initial process capability study: MA/ME/Partnership Quality 8) Packaging: Packaging Team 9) FAIR: MFP 10.1) Part marking: DPMS 10.2) Production Process Run: MFP	Producers: Industrialization Project Manager Partners: Partnership Industrialization Account Manager	Process Owner: PPAP Central IPT OPM	O-PMT O-PMMT
7.4 PPAP	7.4.3 (Last 6 month QN yield >90% AND Interim B) OR (Interim A) OR (Full Approval) is achieved		X	Lead: NPI SM/CMO/GM&T 11 PPAP elements: 1) Design Record: MFP 2) Design Risk Analysis: MA/ME/Partnership Quality 3) PFD: MA/ME/Partnership Quality 4) PFMEA: MA/ME/Partnership Quality 5) Control Plan: MA/ME/Partnership Quality 6) MSA: MA/ME/Partnership Quality 7) Initial process capability study: MA/ME/Partnership Quality 8) Packaging: Packaging Team 9) FAIR: MFP 10.1) Part marking: DPMS 10.2) Production Process Run: MFP	Producers: Industrialization Project Manager Partners: Partnership Industrialization Account Manager	Process Owner: PPAP Central IPT OPM	O-PMT O-PMMT

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7.4 PPAP	7.4.4 Full approval is achieved		X	Lead: NPI SM/CMO/GM&T 11 PPAP elements: 1) Design Record: MFP 2) Design Risk Analysis: MA/ME/Partnership Quality 3) PFD: MA/ME/Partnership Quality 4) PFMEA: MA/ME/Partnership Quality 5) Control Plan: MA/ME/Partnership Quality 6) MSA: MA/ME/Partnership Quality 7) Initial process capability study: MA/ME/Partnership Quality 8) Packaging: Packaging Team 9) FAIR: MFP 10.1) Part marking: DPMS 10.2) Production Process Run: MFP	Producers: Industrialization Project Manager Partners: Partnership Industrialization Account Manager	Process Owner: PPAP Central IPT OPM	O-PMT O-PMMT
7.4 PPAP	7.4.5 Program meeting PPAP target	X		Lead: NPI SM/CMO/GM&T 11 PPAP elements: 1) Design Record: MFP 2) Design Risk Analysis: MA/ME/Partnership Quality 3) PFD: MA/ME/Partnership Quality 4) PFMEA: MA/ME/Partnership Quality 5) Control Plan: MA/ME/Partnership Quality 6) MSA: MA/ME/Partnership Quality 7) Initial process capability study: MA/ME/Partnership Quality 8) Packaging: Packaging Team 9) FAIR: MFP 10.1) Part marking: DPMS 10.2) Production Process Run: MFP	Program Industrialization Manager	Process Owner: PPAP Central IPT OPM	O-PMT O-PMMT
8- Assembly & Test							
PR Element	**PR Deliverable**	**Program**	**POC**	**Responsible**	**Accountable**	**Consulted**	**Informed**
8.1 Assembly & Test Process	8.1.1 All AFS are created to support development engine build and AFS% program target is met.	X		Lead: Development Engineering Partners: Partnership Industrialization Account Manager	Lead: Development Engineering AFS % KPI: Program Industrialization Manager	Process Owner: Engineering A&T Engine Center OPM	O-PMT
8.1 Assembly & Test Process	8.1.2 AFS validation plan to support MA	X		Lead: Development Engineering Partners: Partnership Industrialization Account Manager	Development Engineering	Process Owner: Engineering A&T Engine Center OPM	O-PMT
8.1 Assembly & Test Process	8.1.3 Production AFS are validated and AFS% program target is met.	X		Lead: Development Engineering Partners: Partnership Industrialization Account Manager	Lead: Development Engineering Re-Test ratio KPI: Program Industrialization Manager	Process Owner: Engineering A&T Engine Center OPM	O-PMT
8.1 Assembly & Test Process	8.1.4 Define kanban stock requirements from Partner for first 2 years of engine assembly. Partners to deliver required hardware		X	Lead: Engine Center	Partnership Industrialization Account Manager	Process Owner: Partnership Industrialization IPT OPM	O-PMT
8.2 Assembly & Test Tooling	8.2.1 Development tooling design plan supports FETT	X		Lead: Development Engineering Partners: Partnership Industrialization Account Manager	Development Engineering	Process Owner: Engineering A&T Engine Center OPM	O-PMT
8.2 Assembly & Test Tooling	8.2.2 Production tooling plan aligned with MA engine	X		Lead: Development Engineering Partners: Partnership Industrialization Account Manager	Development Engineering	Process Owner: Engineering A&T Engine Center OPM	O-PMT

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8.2 Assembly & Test Tooling	8.2.3 Production tooling is validated and optimized	X		Lead: Development Engineering Partners: Partnership Industrialization Account Manager	Development Engineering	Process Owner: Engineering A&T Engine Center OPM	O-PMT
8.3 PFMEA	8.3.1 PFMEA completed	X		Lead: Development Engineering Co-Lead: Program PE	Development Engineering	Process Owner: Engineering A&T Engine Center OPM Partnership Quality	O-PMT
8.4 DR's	8.4.1 Develop and execute a plan to close development CAT 1 & CAT 2 DR's	X		AFS&Tooling DR: Development Engineering HW DR: Program PE Operations Lead: OPM	AFS&Tooling DR: Development Engineering HW DR: Engineering program manager	Process Owner: Engineering A&T Engine Center	O-PMT
8.4 DR's	8.4.2 All development CAT 1 & CAT 2 DR's must be closed for DVPM closure	X		AFS&Tooling DR: Development Engineering HW DR: Program PE Operations Lead: OPM	AFS&Tooling DR: Development Engineering HW DR: Engineering program manager	Process Owner: Engineering A&T Engine Center	O-PMT
8.5 Facility Readiness	8.5.1 Partner Assembly Facility ready to support the Production start up	X		Lead: Engine center	Partnership Industrialization Account Manager	Process Owner: Partnership Industrialization IPT OPM	O-PMT
8.5 Facility Readiness	8.5.2 Production facility readiness plan execution on-track to be completed by MA engine	X		Lead: Development Engineering Co-Lead: A&T Engine Center Partners: Partnership Industrialization Account Manager	OPM	Process Owner: Engineering	O-PMT
8.5 Facility Readiness	8.5.3 Production assembly line and test cell ready	X		Lead: Development Engineering Co-Lead: A&T Engine Center Partners: Partnership Industrialization Account Manager	OPM	Process Owner: Engineering	O-PMT O-PMMT
8.6 Performance	8.6.1 Re-test ratio on baseline engine is meeting new program target	X		Lead: Program PE Co-Lead: Development Engineering Operations Lead: A&T Engine Center	Engineering Program Manager	Process Owner: Engineering OPM	O-PMT PMMT O-PMMT
8.6 Performance	8.6.2 Meeting program target for retest ratio	X		Engineering Lead: Program PE Engineering Co Lead: Development Engineering Operations Lead: OPE, A&T Engine Center	Lead: Engineering Program Manager Retest ratio KPI: Program Industrialization Manager	Process Owner: Quality OPM PMMT	O-PMT O-PMMT
8.7 Manufacturing Approval	8.7.1 Manufacturing Approval granted by regulatory authority and meets program intent	X		Lead: OPM	OPM	Process Owner: OPM A&T Engine Center Engineering	O-PMT
8.7 Manufacturing Approval	8.7.2 Partner to provide Conformity Letter for the Current EIS PN at Top Assy PN level	X		Lead: Partnership Industrialization Account Manager Co-Lead: Partnership Quality	OPM	Process Owner: OPM	O-PMT
8.8 OEM support	8.8.2 Consignment inventory hardware list is defined by the IPT and available at OEM		X	Lead: Partnership Industrialization Account Manager Co-Lead: Customer Service	OPM	Process Owner: Customer Service Program PE	O-PMT
8.8 OEM support	8.8.3 Partners Module Tooling is available at the OEM's facility for possible engine maintenance and repair.		X	Lead: Partnership Industrialization Account Manager Co-Lead: Customer Service	OPM	Process Owner: Customer Service Program PE Partnership Office	O-PMT
8.8 OEM support	8.8.4 Publications & Maintenance program issued or on track to be issued at EIS		X	Lead: Customer Service Co-Lead: Partnership Industrialization Account Manager	OPM	Process Owner: Customer Service Program PE Partnership Office	O-PMT
9- Global Chemical Regulations							
PR Element	**PR Deliverable**	**Program**	**POC**	**Responsible**	**Accountable**	**Consulted**	**Informed**
9.1 Global Chemical Regulations	9.1.1 Global chemical assessment completed with risk mitigation plan in place	X		Lead: Global Chemical Regulation Co-Lead: Project Engineering Supply Management: NPI SM/CMO/GM&T	Global Chemical Regulations	Process Owner: RTX Global Compliance OPM	O-PMT
10- Global Trade							

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PR Element	PR Deliverable	Program	POC	Responsible	Accountable	Consulted	Informed
10.1 Global Trade Assessment	10.1.1 Global trade assessment is completed and GT risks are mitigated	X		Lead: Global Trade Professional Specialist	OPM	Process Owner: RTX Global Compliance Partnership Office Partnership Quality & Industrialization Strategic Sourcing	O-PMT

Table 10 – RACI chart

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5.1.3 Program Industrialization Manager (PIM)

The Program Industrialization Manager (PIM) is responsible to develop and deploy the production readiness strategy for a given program. The Program Industrialization Manager is assigned prior IGR A and is responsible for, but not limited to, the following activities:

- Drive for the engine PR Level 2 KPI measurement accuracy and ensure PR KPI targets are met for each IGR and gates as per RACI in section 5.1.2.
- Lead the preparation of the IGR reviews and provide a summary status at each gates including PR exit criteria go/no go status and industrialization next step recommendations.
- Assess Engine level and Producer PR risks and lead the execution of their mitigation plans with involvement of cross-functional teams.
- Maintain PR Deliverable assessment (PWC Form 12248) for the program-type deliverables.
- Represent PR in PMT and other program reviews, e.g. O-PMT, O-PMMT, OPR, …, and raise escalations whenever required.
- Conduct regular reviews with Industrialization Project Manager to ensure the latest communication of program status and strategies.
- Roll up main producer PR risks at the engine level and communicate the risks to OPM.
- Assess and identify POC, and assign them to Industrialization Project Manager.
- Estimate PR budget throughout program execution (e.g. PPAP, ZNC, Resources, Workshops, Travels, etc).
- Assess and report out engine PR status on a monthly basis, or whenever required.
- Support PR process continuous improvement and producer events.
- Work jointly with the Industrialization Project Manager to confirm the PR scope with the producers.
- Coordinate with OPM the Industrialization White Paper.
- Coordinate with Industrialization Project Manager for tracking ZNC top offenders with MA/ME.
- Drive with Industrialization Project Manager the producer's capacity increase plans to secure program ramp up.
- Work with R&T and Industrialization Project Manager to ensure industrialization of New Technologies for TRL lower than 6 .
- Execute program transfer and action closure after IGR D and before IGR E when transfer criteria are met.

5.1.4 Industrialization Project Manager (IPM)

The Industrialization Project Manager is responsible for, but not limited to, the following activities:

- Work with the different teams to closely monitor and support all the industrialization activities with producers to achieve IGR Level from IGR B up to IGR F on its assigned POC.
- Collaborate with internal producer, SM, and Program Industrialization Manager to identify industrialization gaps, define with cross function team a robust mitigation action plan including its tracking until closure and, quantify/escalate risks.

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- Maintain PR Deliverable assessment (PWC Form 12248) for the POC-type of deliverables.
- Closely monitor and ensure achievement of Level 3 KPI targets and exit criteria by gate level as per RACI in section 5.1.2.
- Represent the departmental function when interacting with suppliers, technical specialists and internal program functions, e.g. IPT and other forums.
- Participate into events organized by SM for continuous improvement with internal/external producers and stakeholders .
- Set and lead producer's Production Readiness reviews.
- Under SM guidance, conduct on-site PR reviews at external/internal producers.
- Provide PR support and IMP 07-01-01 / P&WC form 12284 training to external/internal producers.
- Ensure PPAP progress (APQP approach) is timely meeting the Program PPAP targets and coordinate internally with NPI buyers and MA/ME, MFP and other functions responsible for PPAP execution.
- Ensure ZNC progress with external/internal producers and coordinate with MA/ME and Partnership quality.
- Monitor demand planning by gate.
- Follow and monitor NSD (NSE, NSI, NSA) process to meet program planning.
- Track & coordinate industrialization gap, and their mitigation actions and drive to closure.

5.1.5 Partnership Industrialization Account Manager (PIAM)

The Partnership Account Manager is responsible to develop, deploy and execute the production readiness strategy and operations activities for a given program partner. The Partnership Account Manager is responsible for, but not limited to, the following activities:

- Perform and lead Production Readiness with partners by performing regular Production Readiness reviews, PR-L2 and Program Level Events (if needed);
- Accountable for Partners industrial performance and overall Partnership on-site relationship;
- Provide visibility to P&WC on emerging issues on partner location;
- Drive H/W development requirements with partner (as required by the EVP), ensure alignment between operations, engineering and gates;
- Support the development of business processes and procedures in accordance to PR process for the management of partners;
- Drive all operational program requirements as required by the collaboration agreement;
- Monitor Configuration evolution for effective IGR C2 and IGR D;
- Update and maintain PR Deliverables assessment file, including risk cube;
- Act as the single point of contact at the Partner for all program requirements;
- Support engineering change incorporation and retrofits campaign with the Partner (as required)
- Send OPS and PR official communication to partners (Gate assessments, Demand, PPP, and others);
- Support engineering IPTs to understand current issues and to provide visibility of potential risks on the PR milestones, and
- Monitor Export control compliance.

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5.1.6 Partnership Quality (PQ)

Partnership quality is responsible to support the development of the new programs. The activities listed below are performed to ensure the customer requirements are transferred to the partners, the internal processes and best practices are developed to ensure a flawless material flow for the development products and for entry into production. The Partnership Quality is responsible for, but not limited to, the following activities:

- Surveillance of partners: support Quality System & Processes Organization audit plans and audit activities, Canadian Airworthiness Regulation (CAR) follow-up and closure;
- Procert deployment activities: CTQ follow-up, KPCd dashboard, data collection initiation, coordination with ME;
- Support Quality IPT and actions with program partners;
- Support the development of the CQPs/Mgt Plans with program partners: participate to creation, coordination, DRT, ballot, ensure a maintenance plan in place for CQP/Mgt Plans;
- Develop business processes and procedures in accordance to CQPs & Mgt Plans for the management of partners;
- Support the deployment of Intermediate Facilities (IF): IF assessment and identification of the program needs, IF approval and IMP updates, IF process implementation (e.g. Flight Test Quarantine, Material Handling) and IF surveillance (audit);
- Perform internal audit of Industrialization organization procedures, CAR follow-up and closure, prepare and execute internal audit plan;
- Support escape management with partners during the product and process development phases;
- Support and follow-up with partners for ZNC, Procert and PPAP completion
- Develop quality strategies for new programs and participate to advanced industrialization activities (IGR A), and
- Support development of engine programs for quality matters: quality plans with customers and producers, interactions with material flow & integrity to ensure adequate inspection plans for development and test hardware.

5.1.7 Operation Program Engineer/Analyst

Refer (but not limited) to OPMP 3.10, 3.11, and 3.12.

5.1.8 Product Cost Engineer/Analyst

Manage the engine cost information and contribute to company business cases and their evaluation. This cost oversight will influence and guide investment opportunities within Pratt & Whitney Canada thought out the lifecycle of the engine. The Product Cost Engineer/Analyst is responsible for, but not limited to, the following activities:

- Cost leader to guide proposals, conduct financial audits;
- Leading key program stakeholders by establishing cost roll-ups during all phases of production;
- Challenge and recommend various cost savings alternatives via program assumptions and ensure that these recommendations are materialized;

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- Identify and flow down cost goals to engine modules in the Engineering organization as well as to the supply and manufacturing teams in Operations;
- Collaborate with the Operations Program Office to track cost evolution through Production Readiness, Program Management, Partnership, Trade Compliance and Quality teams with respect to cost targets and foresee potential cost headwinds creating a gap to the Financial targets;
- Contribute to a collaborative, innovative and inclusive environment to ensure best practices are applied to all programs and drive continuous improvement, and
- Maintain and improve should cost models by part family to serve a costing best in class information.

5.1.9 Global Chemical Regulation Specialist (GCR Specialist)

The Global Chemical Regulations specialist serve as the primary GCR resource for the Program. GCR is responsible for the compliance with global chemical substances regulatory / customer requirements and risk reduction in product design, sourcing, manufacturing and delivery of products, but not limited to, the following activities:

- Identifies, assesses and communicates GCR risks relevant to materials of concerns (MOC) or substances regulated;
- Identifies, parts with chemical substances and substantiated alternatives under current derogations (authorizations, restrictions) to include engineering changes in new programs budget;
- Identify and communicate potential mitigation plans and actions, if applicable;
- Develops and tracks mitigation actions related to regulation such as advocacy and authorisation, and
- Maintain a database on substance uses to be able to assess risks and the relevant regulatory activities that could have an impact.

5.1.10 GT Focal

The Global Trade Focal serves as the primary GT resource for the Program. The GT Focal is responsible for, but not limited to, the following activities:

- Identify GT risk drivers throughout the Program Lifecycle;
- Determine authorizations requirements based on classifications and sourcing options using the available GT tools;
- Coordinate authorizations requests and availability with GT Operations and GT Professional;
- Escalate risks and ambiguity to the GT Professional, and
- Collaborate with GT Professional and with the appropriate disciplines to develop and manage a global GT risk management plan.

5.1.11 GT Professional

The Global Trade Professional serves as the point of contact between the Program and GT Legal. The GT Professional is responsible for, but not limited to, the following activities:

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- Collaborate with GT Focal and with the appropriate disciplines to develop and manage a global GT risk management plan;
- Advise and resolve policy ambiguity related to the Program;
- Support the development of compliance plans for FAR's and DFARS' that have GT implications, and
- Collaborate with GT Legal and BU.

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6 APPENDICES

6.1 APPENDIX A – PRODUCTION READINESS EXIT CRITERIA AND DELIVERABLES

As detailed in P&WC from 12284, each of the PR deliverables (Column C) are grouped under:

- One of the 10 PR Exit Criteria (column A)
- PR Elements (column B)
- Detailed work instructions to perform the deliverable assessment (column D)
- Applicability to either a Program or a POC (column E and F)
- Program KPI applicability (column G)
- Applicability to one or several IGR (columns H to N)
- R – A – C – I (columns O to R)
- Reference to outside process (column S)
- Data source location (columns T to V)

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6.2 APPENDIX B – PRODUCTION READINESS KPI'S

Program L1 and L2 KPI are displayed into Control tower tab as part of Industrialization PBI

KPI	KPI Level	Source of truth	Calculation	Applicability	Target
MRL%	L1 (all programs)	MRL% in PBI Control Tower	MRL% at IGR date (12 month rolling average)	All Programs with IGR B or more during last 12 months	80%

KPI	KPI Level	Source of truth	Calculation	Applicability	MRL 4	MRL6	MRL7	MRL7	MRL8	MRL9	MRL10
PR EV%*	L2 (program)	PR Deliverable assessment tool	% of applicable PR Deliverable at IGR target AND normalized to PR EV% target	POC	7.0 %	14.4%	24.1%	43.3%	65.8%	87.7%	100%
MRL%	L2 (program)	PR Deliverable assessment tool	% of POC at MRL target	POC			80% MRL7	80% MRL7	80% MRL8	80% MRL9	100% MRL10
NPI to AS % MA to QS %	L2 (program)	SAP NTU Database	% of PN transferred	All BOM (buy parts)						Plan defined	100%
ESA%	L2 (program)	PLM	% of PN meeting ESA target	All BOM with ESA applicability				Program target	Program target	Program target	
Configuration Stability %	L2 (program)	PLM	% of PN without code 11 (purchasing block)	All BOM					Program target	Program target	Program target
LTA %	L2 (program)	CLM Database	% of spend secured under contract (LTA amendment including PN or NPI clause on a signed LTA	All BOM				Program target	Program target	Program target	Program target
Build Ahead %	L2 (program)	SM PBI	% of PN with 28 day Build ahead at supplier in Finish Good or in PWC	PN with contractual build Ahead in LTA						Program target	Program target
Hardware OTD%	L2 (program)	SAP	6 month reconciliated OTD at the PN	All production and experimental Pos						Program target	Program target
Capacity Secured %	L2 (program)	PW Capacity Portal	% of PN with part family risk at low or medium	PN with SDL assigned				Program target	Program target	Program target	Program target
Procert %	L2 (program)	PWC Supplier Portal	% KPCd meeting program Target	PN with KPCd						KPCd with CPK>1	KPCd CPK>1.33
QNLI / Engine	L2 (program)	SAP	Sum of QNLI generated past 6 months divided by number of engines shipped during same period (both Experimental and Production)	All BOM	Program target	Program target	Program target	Program target	Program target	Program target	Program target
PPAP %	L2 (program)	SAP NTU database	% PN-VC meeting program target	Applicable PPAP PN-VC				PPAP Plan	Interim B before prod PO HW date	Interim A or QN yield > 90%	Full Approval
AFS%	L2 (program)	ECATES	% of AFS meeting program IGR target	Engine				Created		Validated	
Re-Test Ratio	L2 (program)	Engine Center Test database	Number of production engine tested during last 3 months divided by number of engines delivered during same period.	Engine						Program target	Program target
QNLI/Part	L3 (Part)	SAP	Sum of QNLI generated past 6 months divided by number of parts received during same period (both Experimental and Production)	POC						< 0.1 90%+ QN yield)	0

* PR Earned Value target % by GIR is dependant on the number of PR deliverable applicable for each IGR as per from 12284. If PWC form 12284 changes after the implementation of this IMP 07-01-01 document, the form PR Earned Value target gets precedence.

Table 11 – Production Readiness KPI applicability

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